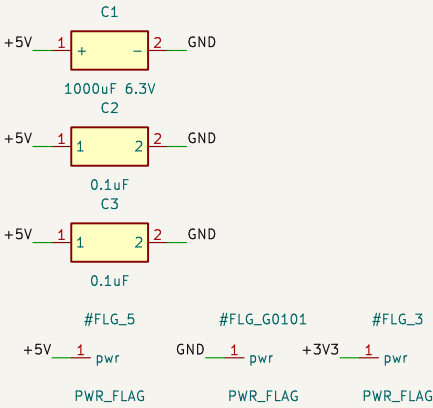
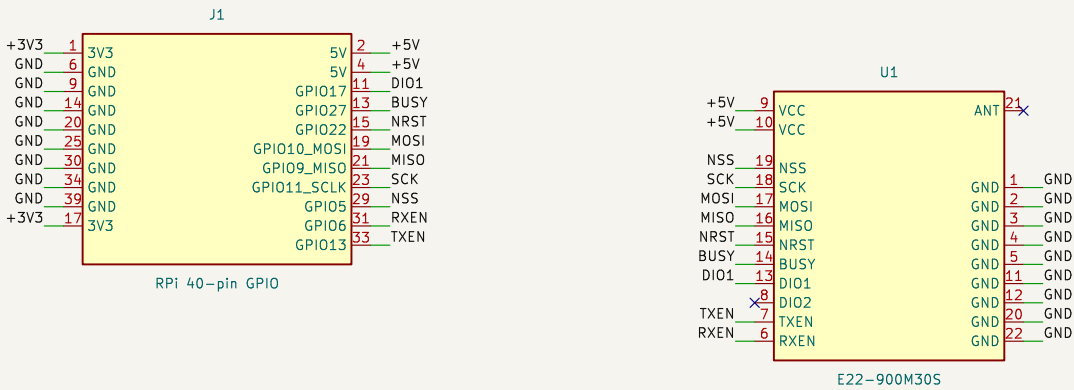


MeshGW Pi HAT – E22–900M30S carrier (rev A DRAFT, 2026–08–12)

DO NOT ORDER: E22 land pattern is datasheet–derived and has NOT been checked against a physical part. That is now the ONLY board blocker – HAT mechanical geometry was verified vs the Pi 3B+ on 2026–08–12.



POWER: E22 VCC = +5V (Pi header pins 2/4), NOT 3V3.

Datasheet: 2.5–5.5V range but ‘>=5.0V ensures output power’, 650 mA instantaneous on a TX burst. The Pi 3V3 regulator must not carry that. Logic level is 3.3V so SPI wires straight to the Pi with no level shifting.

RF: pad 21 (ANT stamp hole) is NO–CONNECT by design. The antenna leaves through the module’s own IPEX connector via an IPEX–>SMA bulkhead pigtail. Nothing on this 2–layer board is impedance–controlled, and a bad RF trace passes every automated check.

TXEN/RXEN are REQUIRED – the E22 has an RF switch. Leave them unconnected and the PA transmits into a closed switch.

NEVER power the module without an antenna: 1W into an open circuit damages the PA.

PIN MAP (BCM) – generated from netlist.py, the ONE source:

NSS GPIO5 (hdr 29) BUSY GPIO27 (hdr 13)
SCK GPIO11 (hdr 23) DIO1 GPIO17 (hdr 11)
MOSI GPIO10 (hdr 19) NRST GPIO22 (hdr 15)
MISO GPIO9 (hdr 21) TXEN GPIO13 (hdr 33)
RXEN GPIO6 (hdr 31)

This sheet and the PCB are both generated from it, so they cannot drift. config/meshtasticd/config.yaml is NOT generated – re–pin here and you must hand–edit it or the radio won’t start. Diverges from MeshAdv–Pi–Hat deliberately; netlist.py says why.

GEOMETRY: 65 x 56.0 mm, R3 corners, notched for the Pi 3B+ PoE header J14 (those pins can carry ~48–57V on a PoE switch). Verified vs the official HAT spec 2026–08–12; check_geometry.py asserts 27 of those facts against the ROUTED board every run.

Empire12

Sheet: /
File: meshgw–hat.kicad_sch

Title: MeshGW Pi HAT – E22–900M30S carrier

Size: A3 Date: 2026–08–12

KiCad E.D.A. 9.0.8

Rev: A

Id: 1/1